MIL-C-24643/38B
3 October 1986
SUPERSEDING
MIL-C-0024643/38A(SH)
8 November 1985
MIL-C-24643/38
28 September 1984

MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, 600 VOLTS, TYPE LSECM (INCLUDING VARIATION LSECMA)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the cable described herein shall consist of this specification and the latest issue of MIL-C-24643.

This cable can be manufactured in two variations: LSECM (unarmored) and LSECMA (armored).

REQUIREMENTS:

Qualification required.

Construction (watertight)

First - Copper conductor, in accordance with ASTM B 286, size 20-7, tin coated. Maximum diameter 0.039 inch.
Second - Separator may be used at manufacturer's option where required to provide free-stripping insulation.
Third - Cross-linked polyethylene insulation. Nominal thickness 0.020 inch. Colored insulation - one black and one white conductor for each pair.
Fourth - Two conductors, one black and one white, cabled together with a nominal lay of 2.5 inches to form a pair.
Fifth - Binder tape over each pair, at manufacturer's option, applied helically with overlap.
Sixth - Braided shield of AWG no. 34 or no. 36 tin coated copper. Braid angle of 30 to 35 degrees; minimum coverage of 85 percent.
Seventh - Shield insulation of two polyester tapes, type G in accordance with MIL-I-631, sealed. Standard identification code applied by method 2, on inner tape.

(B) denotes changes.

AMSC N/A FSC 6145
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Groups of 7 (8 groups)

First - Copper conductor, AWG no. 18, class B stranding, (7 by 0.0152 inch), tin coated. Maximum diameter 0.050 inch.
Second - Ethylene propylene rubber or cross-linked polyethylene insulation. Nominal thickness 0.008 inch. Colored insulation - five black and two white conductors for each group.
Third - Seven conductors, five black and two white, with the two white conductors adjacent in the layer of six, cabled with a lay not greater than 2 inches, to form each group.
Fourth - Two polyester tapes, type G in accordance with MIL-I-631, sealed. Standard identification code applied by method 2, on inner tape.

ASSEMBLY:

First - Five pairs cabled to form a core. Three pairs and eight groups of seven cabled over the core, with a lay not greater than 24 times the pitch diameter of the layer. Pair nos. 1 through 5 in core; the sequence of the outer-layer - pair no. 6, group nos. 1, 2 and 3; pair no. 7; group nos. 4, 5 and 6; pair no. 8; group nos. 7 and 8. Fillers shall be employed to form a well-rounded assembly.
Second - Binder tape applied helically with overlap.

Construction variant for type LSECM

Third - Cross-linked polyolefin jacket. Nominal thickness 0.100 inch. Cable surface marking required.

Construction variant for type LSECMA

Third - Cross-linked polyolefin jacket. Nominal thickness 0.100 inch.
Fourth - Braided metal armor and paint.

NOTE: 1. Variant LSECM completed cable diameters: minimum 1.290 inches, maximum 1.370 inches. Military part number M24643/38-01UN.
2. Variant LSECMA completed cable diameters: minimum 1.340 inches, maximum 1.420 inches. Military part number M24643/38-02AN.

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EXAMINATION AND TESTS:

	Requirements
Basic electrical:	
Conductor resistance - ohms/1000 feet, at 25°C, maximum	
ASTM B 286, size 20-7	11.03
AWG no. 18 (class B)	7.47
Voltage withstand - volts, root mean square, minimum	
Conductor to conductor	2000
Conductor to shield	1000
Shield to shield	500
Insulation resistance	
Megohms/1000 feet, minimum	500
Conductor continuity	No failure
Group A:	
Visual and dimensional	No failure
Watertightness - see MIL-C-24643 for limits of water leakage	No failure
Capacitance	
Mutual capacitance - at 1 kHz, pF/ft, maximum	30
Group B:	
Drip - 95 ± 1°C	Zero
Tear - pounds per inch thickness, minimum (ASTM D 470)	35
Physicals (unaged)	
Insulation	
Ethylene propylene rubber	
Tensile strength - lb/in^2, minimum	700
Elongation - percent, minimum	250
Cross-linked polyethylene	
Tensile strength - lb/in^2, minimum	1800
Elongation - percent, minimum	250
Jacket (cable)	
Tensile strength - lb/in^2, minimum	1300
Elongation - percent, minimum	160
Group C:	
Physicals (aged)	
Insulation	
Ethylene propylene rubber	
Air oven	
Tensile strength - percent of unaged, minimum	75
Elongation - percent of unaged, minimum	75
Cross-linked polyethylene	
Air oven	
Tensile strength - percent of unaged, minimum	80
Elongation - percent of unaged, minimum	80

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EXAMINATION AND TESTS: (Continued)

	Requirements
Jacket (cable)	
Air oven	
Tensile strength - percent of unaged, minimum	60
Elongation - percent of unaged, minimum	60
Hot oil immersion	
Tensile strength - lb/in^2, minimum	50
Elongation - percent, minimum	50
Shrinkage	No failure
Permanence of printing (jacket) type LSECM only - cycles, minimum	250
Cable filler removability	No failure
Heat distortion - percent of unaged, maximum	30
Armor - conformance to material, construction and coverage (type LSECMA only)	No failure

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B and C, plus the following:

	Aging and compatibility (cable) (125 ± 5°C)	No failure
(B)	Abrasion resistance (jacket) - scrapes, minimum	75
	Acid gas equivalent - percent, maximum	
	Jacket	2
	Fillers	2
	Insulation	18
	Halogen content - percent, maximum	
	Jacket	0.2
	Fillers	0.2
	Flame propagation (cable)	No failure
	Immersion (jacket)	
	Tensile strength - percent of unaged, minimum	50
	Elongation - percent of unaged, minimum	50
	Smoke index, maximum	
	Jacket	25
	Fillers	—
(B)	Insulation	45
	Toxicity index, maximum	
	Jacket	5
	Fillers	5
	Insulation	1.5
	Durometer (jacket) - (type A) hardness, minimum	80
	Weathering (jacket)	No failure

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UNIT ORDERING LENGTH: 1000 feet (nominal)

NOTE: Not for Air Force use.

Custodians:
Army - MI
Navy - SH

Review activities:
Army - AV, CR, ER
Navy - EC
DLA - IS

User activities:
Army - ME, AR, AL
Navy - CG

Preparing activity:
Navy - SH
(Project 6145-1005-38)

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